THREE EQUIVALENTS OF A SCHEME HAVING AFFINE DIAGONAL: AFFINE PAIRWISE INTERSECTIONS OF AFFINE OPENS, OR EXISTENCE OF AN OPEN COVERING WHOSE ALL FINITE INTERSECTIONS ARE AFFINE; IN PARTICULAR ANY SEPARATED SCHEME SATISFIES THEM.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 435-447 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 449-460, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33g.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01S8: sources/stacks-coherent/pinned-original-source.tex, source lines 460–460. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01KP: sources/stacks-coherent/pinned-original-source.tex, source lines 460–460. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a scheme. The following are equivalent*

(1) *X has affine diagonal $\Delta : X \to X \times X$,*
(2) *for $U, V \subset X$ affine open, the intersection $U \cap V$ is affine, and*
(3) *there exists an open covering $\mathcal{U} : X = \bigcup_{i \in I} U_i$ such that $U_{i_0 \ldots i_p}$ is affine open for all $p \geq 0$ and all $i_0, \ldots, i_p \in I$.*

In particular this holds if X is separated.

Proof. Assume X has affine diagonal. Let $U, V \subset X$ be affine opens. Then $U \cap V = \Delta^{-1}(U \times V)$ is affine. Thus (2) holds. It is immediate that (2) implies (3). Conversely, if there is a covering of X as in (3), then $X \times X = \bigcup U_i \times U_{i'}$ is an affine open covering, and we see that $\Delta^{-1}(U_i \times U_{i'}) = U_i \cap U_{i'}$ is affine. Then Δ is an affine morphism by Morphisms, Lemma 01S8. The final assertion follows from Schemes, Lemma 01KP. □

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